# Fall 2026 – Hack-on-the-Hill
# 3rd International Hackathon
## Problem Statements and Judging Criteria

Select one of the challenges proposed in the document and develop an AI-driven solution and prototype.

## Team Instructions

Each participating team must select one problem statement from this document and propose an AI-driven solution. You may make reasonable assumptions, but these must be clearly justified in your presentation.

You are expected to:

- Clearly identify the intended users, their needs, and the practical consequences of the problem.
- Design a coherent system with useful, innovative features and a feasible scope for the hackathon.
- Incorporate appropriate AI techniques and explain the value they add compared with a simpler approach.
- Justify design decisions, data sources, assumptions, and trade-offs, including privacy, accessibility, and human oversight.
- Build a working prototype demonstrating core functionality from input through processing to output.
- Test representative examples and at least one difficult or failure case. Report results honestly, including limitations.
- Prepare a clear 10-minute presentation that explains the problem, system design, AI approach, evaluation, and working demonstration.

### Prototype expectations

A focused prototype is sufficient; a production-ready system is not required. Public, permissioned, or clearly labeled synthetic data may be used. Identify simulated components, external models, reused code, and AI development tools. Explain what your team built and demonstrate the implemented workflow rather than relying only on slides or a single prompt.

Use the design questions to guide your decisions; they are not a checklist of mandatory features. Teams should choose a manageable scope and explain with reasoning about the features kept out of the prototype scope.

All submissions will be evaluated using the same judging criteria below.

# Judging Criteria

| Criterion | Description | Weight |
|---|---|---|
| Problem Understanding | Clearly defines users and the problem; explains practical constraints and challenges; establishes a plausible benefit and a way to measure it. | 20% |
| Solution Design and Innovation | Presents a coherent, feasible system design; offers a useful improvement over obvious or existing approaches; addresses usability, accessibility, privacy, and relevant risks through design choices. | 20% |
| AI Usage and Appropriateness | Justifies the AI method and data choices; demonstrates its contribution using test cases or a simple baseline; explains uncertainty, limitations, and appropriate human review. | 20% |
| Presentation and Explanation | Communicates the problem and solution clearly; gives an understandable demonstration; explains decisions, results, limitations, and answers questions accurately. | 20% |
| Prototype and Functionality | Demonstrates a working input-to-output workflow; implements the stated core features and handles a relevant failure case. | 10% |
| Collaboration | Teams with members from at least two different institutions receive the full 10%. Teams whose members are all from one institution receive 0% for this criterion. | 10% |

## Scoring guidance

For each criterion other than collaboration, award 0% of its available points when absent, 25% for minimal evidence, 50% for partial evidence, 75% for strong evidence with minor gaps, and 100% for convincing evidence. Award collaboration points according to the institutional eligibility rule above. Add the criterion scores and average judges' totals. Judge prototype operation under functionality and clarity of the demonstration under presentation.

Credit demonstrated benefits and honest testing; do not require large models, paid services, or production deployment. Disclosed synthetic inputs are acceptable, but simulated outputs do not demonstrate working functionality.

## List of Problems


- Problem 1: Human Solvable Puzzles That Challenge AI
- Problem 2: Disaster Relief Request and Resource Coordination
- Problem 3: When should an AI Assistant Stop and Ask?
- Problem 4: ScamShield AI: An Explainable, Multilingual Defense Against AI-Generated Scams and Phishing

# Problem 1: Human Solvable Puzzles That Challenge AI

## THE PROBLEM

Online assessments and reasoning benchmarks need fresh tasks that measure more than memorized answers. However, a puzzle that seems difficult for one AI model may be easy for another, and making puzzles obscure can also make them unfair or inaccessible to people. Simply asking a model to create a difficult puzzle does not establish that the puzzle is valid or that humans can solve it.

Build an AI agent that generates puzzles, verifies their solutions, tests them against selected AI systems, and refines them using the results. The goal is to demonstrate a measurable human–AI performance gap on a clearly defined puzzle family, not to claim that any puzzle is permanently AI-proof.

## DESIGN CONSIDERATIONS

- What puzzle family will you target, and what makes it useful for education, reasoning evaluation, or assessment?
- How will you verify that each puzzle is solvable, that its rules are consistent, and that its expected answer is correct?
- How will the agent generate, test, and revise puzzles while avoiding an endless loop or excessive model costs?
- How will you compare human and AI performance fairly using defined instructions, time limits, attempt counts, and permitted tools?
- How will you separate examples used to refine the generator from unseen examples used for final evaluation?
- How will you account for accessibility, language, prior knowledge, and the limitations of a small volunteer sample?
- How will you report results for the models actually tested and avoid claiming the puzzles are secure authentication mechanisms?

# Problem 2: Disaster Relief Request and Resource Coordination

## THE PROBLEM

After a flood, storm, or other local disruption, requests for assistance may arrive through different channels, such as messages, forms, and calls in different languages. Reports may be duplicated, outdated, or incomplete. The volunteers and relief groups must match limited supplies, transport, and shelter capacity to the real and changing needs. A useful coordination tool that recognize uncertainty and allow people to review consequential decisions could offer great help.

Design an AI-assisted system that converts incoming reports into structured requirements, identifies their validity and likely duplicates, and proposes feasible matches to available resources. Demonstrate a bounded scenario, such as coordinating supply deliveries across a few neighborhoods, with simulated requests and resource updates.

## DESIGN CONSIDERATIONS

- How will AI extract needs, location, urgency indicators, and missing information from inconsistent or multilingual reports?
- How will the system distinguish duplicate reports from separate households at the same location without deleting valid requests?
- How will it match needs to resources while respecting capacity, delivery windows, accessibility requirements, and travel constraints?
- How will recommendations change when a resource is exhausted, a road becomes unavailable, or a request is corrected?
- How will coordinators inspect evidence, override suggestions, and approve assignments before any action is taken?
- How will you protect personal information and prevent unverified reports or unequal digital access from distorting priorities?
- How will you compare the prototype with a simple first-come-first-served process using simulated fulfillment rates, delay, and unmet needs?

# Problem 3: When should an AI Assistant Stop and Ask?

## THE PROBLEM

AI assistants are moving from answering questions to taking actions. They book appointments, reorder groceries, pay bills, refill prescriptions and adjust smart-home devices on a person's behalf. This is helpful until the assistant acts on a wrong assumption, a misread message, or an instruction slipped in by someone else. Asking the person before every action makes the assistant useless. Never asking makes it dangerous. Design the checkpoint for an AI assistant that serves a specific group, for example an older adult living alone and a family caregiver, a small-business owner, or a student managing money for the first time.

## DESIGN CONSIDERATIONS

- Which actions may the assistant take on its own, which need quick confirmation, and which must always go to a person, and why?
- What evidence does the assistant show to justify an action? "Your pharmacy texted that the refill is ready" is different from "I think you need a refill."
- How do you avoid confirmation fatigue, where people approve requests without reading them?
- What happens when the person is unavailable? Does the assistant wait, choose a safe default, or contact someone else?
- How could a scammer try to trick the assistant into acting, and how does your design stop it?
- Can the person see afterward what the assistant did and why, and who is responsible when it gets something wrong?
- How will you test the checkpoint safely and show that it works? Use invented users and sample data only, and compare it with an assistant that always asks and one that never asks, on scripted requests that include at least one scam attempt.

# Problem 4: ScamShield AI: An Explainable, Multilingual Defense Against AI-Generated Scams and Phishing

## THE PROBLEM

AI-generated scams and phishing increasingly target students and the general public through email, messaging apps, and QR codes. Most existing detection tools are either too technical for a general user or provide binary verdicts with no explanation, leaving users unable to judge trustworthiness or act appropriately, particularly for elderly and other vulnerable users. By narrowing to text-based and link/QR-based vectors, teams can go deep on explainability and usability rather than spreading thin across every media type.

Teams will build a system that detects and explains AI-generated scams and phishing attempts, focused specifically on suspicious messages, emails, links, QR codes, and selected media evidence, rather than attempting to cover every form of image, video, audio, and website manipulation. This narrower scope keeps the challenge realistic and demo-ready within a short international hackathon while still addressing a high-impact, high-frequency threat.

## DESIGN CONSIDERATIONS

- How will you detect likely scams, phishing attempts, impersonation, and manipulated content in messages, emails, links, and QR codes?
- How will your explainable risk-scoring engine identify specific warning signs rather than provide only a verdict?
- How will safe-handling logic recommend actions without automatically opening dangerous links?
- How will multilingual support and accessibility needs be reflected in both detection and plain-language explanations?
- How will you research real-world scam and phishing patterns affecting students and vulnerable users?
- How will you design non-alarming warnings that encourage safe action without causing alert fatigue, and build an awareness or prevention campaign?
- What escalation and reporting pathways will you define for campus IT, security, or relevant authorities?
- How will IT/CS and Management/Business team members contribute to detection, risk communication, outreach, and governance, and demonstrate protective impact and real-world adoption in the final pitch?